

AI Research Outline

I. Artificial Intelligence Research

A. Capability Research

1. Foundation Models
 - a. Architectures
 - b. Scaling Laws
 - c. Training Methods
 - d. Data Curation
 - e. Multimodal Systems
2. Reasoning Systems
 - a. Chain-of-Thought Methods
 - b. Search
 - c. Tool Use
 - d. Planning
 - e. Agents
3. Reinforcement Learning
 - a. RLHF
 - b. RLAIIF
 - c. Online Reinforcement Learning
 - d. Self-Play
4. Efficiency Research
 - a. Quantization
 - b. Distillation
 - c. Inference Optimization
 - d. Hardware–Software Co-Design

B. Evaluation and Measurement

1. Benchmarking
2. Robustness Testing
3. Capability Forecasting

C. Alignment and Safety

1. Preference Learning
2. Control Methods
3. Governance
4. Transparency and Interpretability

D. AI Science

1. Learning Theory
2. Optimization Theory
3. Scaling Laws
4. Representation Theory
5. Mechanistic Interpretability

II. Interpretability Research

A. Behavioral Interpretability

1. Black-Box Analysis
2. Behavioral Probing

B. Representation Analysis

1. Information Localization
2. Representation Structure

C. Mechanistic Interpretability

1. Circuit Analysis
2. Algorithm Discovery
3. Causal Analysis
4. Feature Analysis

D. Neuroscience-Inspired Interpretability

1. Representational Similarity
2. Dynamical Systems Analysis
3. Population Coding
4. Computational Neuroscience Comparisons

III. Mechanistic Interpretability Within the AI Landscape

A. Relationship to Capability Research

1. Understanding Learned Computation
2. Not Primarily Focused on Improving Performance

B. Relationship to Alignment Research

1. Transparency
2. Monitoring
3. Safety Applications

C. Relationship to AI Science

1. Reverse Engineering Neural Networks
2. Discovering Computational Structures
3. Explaining Model Behavior Mechanistically

D. Major Contemporary Schools

1. Circuit-Centric Approaches
 - a. Transformer Circuits
 - b. Reverse Engineering Algorithms
 - c. Computational Graph Analysis
2. Feature-Centric Approaches

- a. Sparse Autoencoders
- b. Feature Dictionaries
- c. Neural Ontology Mapping

IV. Central Question of Mechanistic Interpretability

- A. Identification of Internal Computations
- B. Discovery of Learned Algorithms
- C. Mapping of Representational Structures
- D. Explanation of Model Behavior Through Internal Mechanisms
- E. Understanding What Computational Structures Gradient Descent Creates